

Juspay - Company Briefing & Interview Experience Summary

1. Round 1: Problem-Solving on Maze Traversals

- **Problem 1:** Optimal path traversal from source to destination in a maze with multiple entries and one exit.
 - **Problem 2:** Finding the shortest distance between two values in the maze after graph construction using DFS/BFS.
 - **Problem 3:** Identifying all possible paths between nodes to determine optimal entry/exit strategies.
 - **Evaluation Criteria:**
 - Time complexity (should ideally be better than $O(\log n)$ or $O(n)$).
 - Correct use of algorithms.
 - Code efficiency and scalability.
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2. Round 2: Hackathon Part A – Tree of Space

- **Task:** Solve a complex problem divided into three sub-problems involving tree data structures.
 - **Requirements:**
 - Build a tree from a set of values.
 - Apply various traversals (level-order, in-order, pre/post-order) as per the rules.
 - Perform node-specific operations (e.g., locking, incrementing values).
 - Maintain efficiency and correct time complexity.
 - **Key Considerations:** Balanced tree creation, traversal algorithms, adherence to constraints.
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3. Round 3: Hackathon Part B – Multithreaded System Implementation

- **Continuation of "Tree of Space" problem:**
 - Present and explain the solution and logic from Part A.
 - Implement a multi-threaded approach to the same problem.
 - Handle race conditions and concurrency issues at the hardware level.

- Ensure extensible and thread-safe code.
 - **Format:** One-on-one mentorship and evaluation by an interviewer throughout the process.
 - **Duration:** Full-day monitored session.
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4. Final Round: Technical + HR

- **HR Component:** Basic personal and resume-related questions.
 - **Technical Component:**
 - Core CS subjects: Operating Systems, Computer Networks, System Design, etc.
 - Topics include multithreading, multiprocessing, secure sockets (SSL), and logical problem-solving.
 - Emphasis on deep understanding over surface-level knowledge.
 - System Design: Real-world scenarios requiring end-to-end architectural breakdowns.
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Additional Note:

- **Work Experience Insight:** Experience with **functional programming**, which contrasts with the object-oriented paradigm commonly taught in academia. Emphasis on learning function-centric design for success in Juspay's environment.